

DiffuView: Multi-View Diffusion Pretraining for 3D-Aware Robotic Manipulation

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Abstract

Robotic manipulation from visual observations remains challenging due to the lack of 3D consistent representations that can generalize across diverse viewpoints and sensor configurations. Existing methods, primarily based on masked autoencoders or neural scene representations, struggle to capture robust view correspondences due to a lack of global 3D consistency or the requirement for scene specific optimization. Crucially, while multi-view diffusion models have recently shown tremendous success in 3D aware generative synthesis, their powerful representations offer a promising direction for achieving viewpoint robust visuomotor control. In this paper, we introduce *DiffuView*, a novel framework that learns unified 3D aware representations through multi-view diffusion pretraining and deploys them for imitation learning. Specifically, *DiffuView* models the conditional generation of target views given source observations within a diffusion framework, enabling the network to implicitly recover scene geometry and enforce view consistency. The pretrained diffusion network is then utilized as a powerful visual backbone for an action policy, allowing robust control under varying viewpoints and visual conditions. We evaluate *DiffuView* on two challenging benchmarks, *MetaWorld* and *Libero*. Extensive experiments in both simulation and realworld scenarios demonstrate that *DiffuView* achieves superior generalization, improving success rates under viewpoint shifts by nearly 20% compared with existing methods.

1. Introduction

The traditional paradigm of Visual Behavior Cloning primarily focuses on extracting informative features from visual input to generate robot actions. In recent developments, researchers have introduced language-conditioned modulation and expressive generative modeling to further improve this mapping process [3, 6, 25, 34, 43, 65], where the incorporation of language serves as a pivotal mechanism for

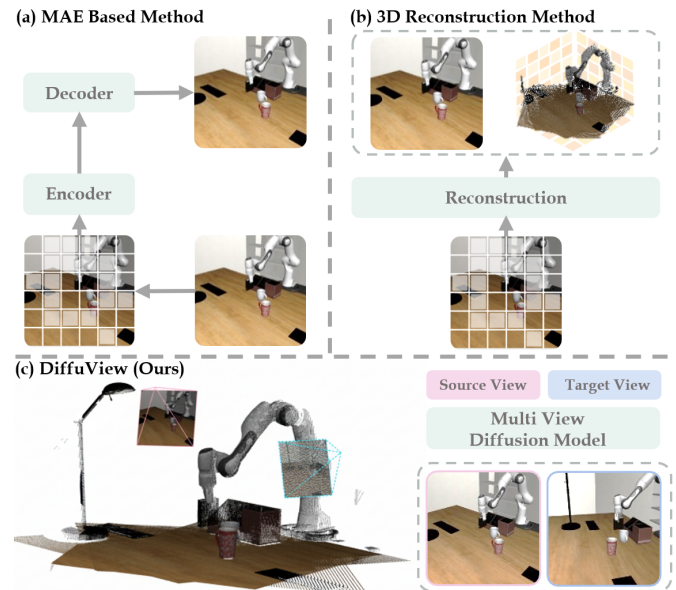


Figure 1. Comparison of visual representation learning paradigms for robotic manipulation. (a) MAE based methods learn visual representations by reconstructing masked regions from observations. (b) 3D reconstruction methods lift 2D observation features into implicit 3D latent spaces through neural rendering. (c) Our method leverages a multi view diffusion model that learns 3D consistent and geometry aware representations by generating novel target views conditioned on source observations and camera poses, enabling unified and view robust visual understanding for downstream manipulation tasks.

achieving robust multi task generalization.

Although end-to-end approaches hold great promise for achieving generalizable robotic control, progress remains limited by the scarcity of large-scale and diverse datasets containing paired visual and action annotations. To overcome this data bottleneck, recent studies have turned to leveraging advances in computer vision, particularly self-supervised and large-scale visual pretraining, to obtain transferable representations that can benefit downstream policy learning [32, 35, 52]. However, most of these approaches rely solely on 2D imagery, lacking awareness of

the underlying 3D spatial structure that is critical for embodied interaction [21, 38, 64]. Inspired by this, pretraining models on multi-view data to acquire 3D-consistent representations emerges as a key step toward more robust and generalizable visuomotor learning.

In prior pretraining methods, masked autoencoding based approaches have become particularly popular, as illustrated in Fig. 1(a). Representative works such as MVP [52], 3D-MVP [38], LIFT3D [21], Embodied-MAE [10], and CL3R [7] adopt this paradigm, where the model learns to reconstruct masked observations according to a probabilistic objective of the form:

$$p_{\theta}(\hat{O}_i | O_i[m]), \quad (1)$$

with O_i representing input observations, m representing binary mask. With the rapid advancement of neural rendering, several subsequent works such as GNFactor [58], SPA [64], and PDFactor [48] have leveraged its capability to lift 2D representations into 3D latent spaces. This process can be formulated as learning a probabilistic mapping:

$$p_{\theta}(V | O_i, P_i), \quad \text{with } \hat{O} = f_{\phi}(V, P), \quad (2)$$

where P_i represents the camera pose, V is the latent 3D representation, including triplane features [48, 54], volumetric grids [55, 64], or Gaussians fields [53, 61]. And f_{ϕ} is a neural rendering function, providing a continuous and spatially structured scene embedding. These pretrained 3D-aware representations have proven effective in supporting downstream action learning for robotic manipulation.

However, these methods still struggle to learn a unified 3D representation across different viewpoints or sensing modalities, limiting their robustness and generalization. Motivated by recent progress in diffusion-based implicit 3D novel view synthesis, such as Free3D [60], Cat3D [14], ViewCrafter [57], Gen3C [41], and Bolt3D [47], we leverage a Multi View Diffusion Model(MVDM) to address this limitation. Specifically, we model the probabilistic generation of a target view $\hat{O}_j$ conditioned on a source observation O_i and their corresponding camera poses P_i and P_j as:

$$p_{\theta}(\hat{O}_j | O_i, P_i, P_j). \quad (3)$$

By learning this conditional distribution, the model captures cross-view relationships and encodes 3D consistent representations within a unified diffusion framework, resulting in more robust and view consistent visual features for downstream manipulation tasks.

To this end, we propose DiffuView, a diffusion based framework for imitation learning that aims to acquire 3D consistent visual representations for robotic manipulation. Our method consists of two stages, as illustrated in Fig. 2. In the first stage, we utilize a multi-view diffusion model [4]

as a starting point. This model is capable of reconstructing target views from source observations, having been pretrained on large-scale, web-crawled multi-view datasets including Objaverse [9], Co3Dv2 [40], Scannet [8], Real10k [63] and so on. Subsequently, we fine-tune the model on a curated robot centric dataset that integrates synthetic data with real world robot demonstrations. This process bridges the gap between general vision and embodied control, and encourage the pretrained model to learn unified, 3D-aware representations through the capture of spatial correspondences across multiple viewpoints. Notably, while multi-view information is leveraged during training, the resulting model requires only a single-view observation during deployment. In the second stage, we integrate the pretrained model into an action diffusion policy network, where the learned representations serve as structured visual features to guide action generation. The policy is trained via imitation learning on robotic manipulation trajectories, resulting in robust visuomotor control that generalizes across varying camera poses and visual conditions. Overall, DiffuView demonstrates that enforcing 3D consistency in representation learning effectively enhances both the transferability and generalization of diffusion based manipulation policies. To summarize, our contributions are as follows:

- We propose DiffuView, a novel diffusion-based representation learning framework for robotic manipulation that learns 3D consistent visual representations through multi-view diffusion pretraining and flexible view deployment.
- We integrate the pretrained diffusion representation into an imitation learning policy, demonstrating that enforcing 3D consistency significantly enhances visuomotor robustness and generalization across varying camera poses and visual conditions.
- We conduct extensive experiments in both simulation and realworld robotic manipulation scenarios, showing that DiffuView achieves superior visual robustness and transfer performance compared to existing approaches.

2. Related Works

2.1. Robotic representation learning

Robotic representation learning builds on large scale visual foundation models. Generic self supervised objectives such as contrastive learning, self-distillation, mask autoencoding, diffusion models, together with language-image alignment, produce strong 2D features that can be adapted to control [13, 17, 36, 39]. Building on these advances, a line of work learns robot-specific 2D embeddings from human videos or multitask manipulation data [22, 31, 33, 35, 52], but purely image-based encoders often lack 3d information. To address this, SPA [64] augments ViT with neural rendering to inject 3D spatial awareness, while Lift3D [21] lifts 2D foundation models into depth space and for policy learn-

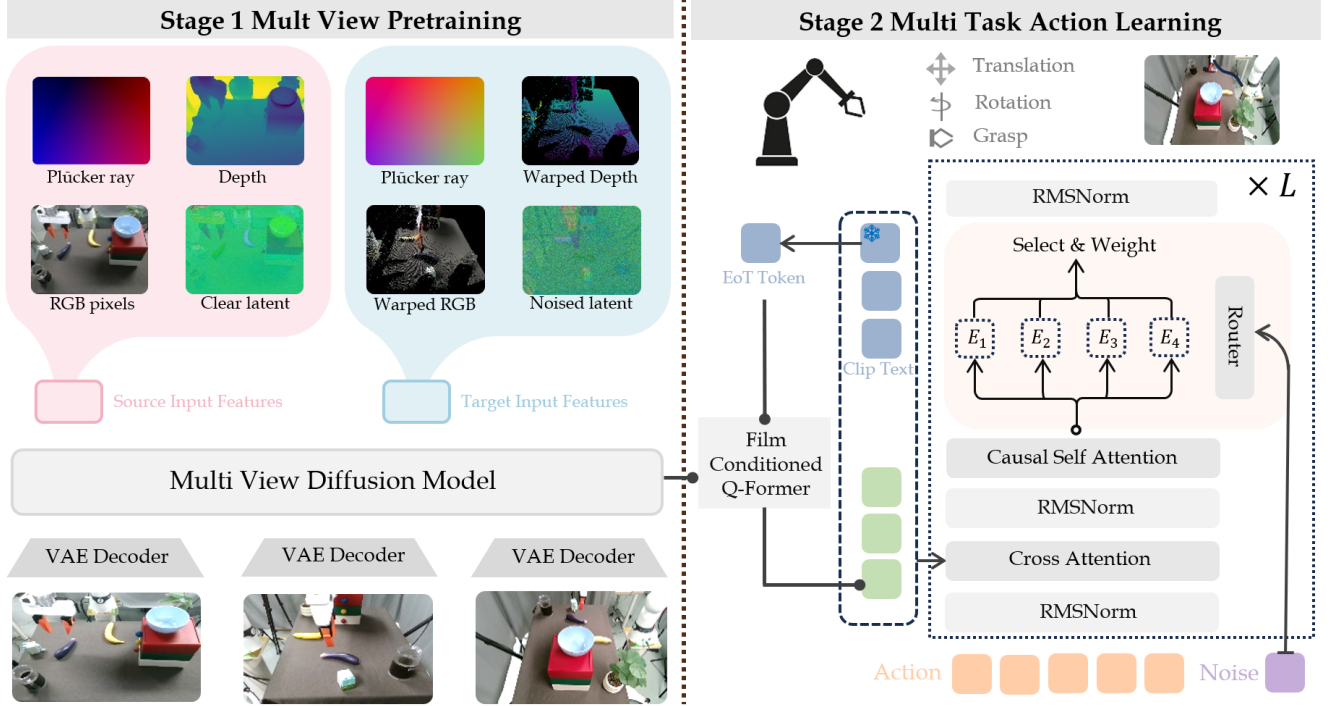


Figure 2. (Left) **Stage 1:** Multi-view diffusion pretraining reconstructs target views from source observations using Plücker rays, depth, and warped RGB depth pairs to learn 3D consistent latent features. (Right) **Stage 2:** The pretrained backbone serves as a 3D aware encoder; a FiLM conditioned Query Transformer (Q-Former) with CLIP text guidance produces task aware tokens that condition a Mixture of Denoising Experts (MoDE) diffusion policy. A noise token $\eta(\sigma_t)$ modulates attention and drives a noise conditioned router that sparsely activates experts, enabling faster inference while maintaining denoising performance.

ing. Complementarily, EmbodiedMAE [10] pretrains a unified RGB-depth-point cloud masked autoencoder. Together, these works suggest that combining pretraining with 3D geometry is key to robust and generalizable robotic control.

2.2. Multi task action learning

Multi task action learning, where a single agent acquires and executes a variety of tasks under a unified policy or framework, has long been studied in the robotics communities. Recent advances in multi task action learning for robotics have progressed along three aspects: data scaling, architecture design, and representation grounding. On the data side, large cross-robot corpora enable generalist policies that transfer across embodiments and tasks, including Open X-Embodiment [37], DROID [24] and so on. Architecturally, transformer based policies have become the most widely used backbone for unifying diverse observations and behaviors [1, 3, 16, 46]. In parallel, diffusion based controllers demonstrate strong closed-loop performance and compositionality [6, 18, 42, 43]. A complementary thread grounds multi task policies in vision-language-action (VLA) priors, leveraging internet-scale knowledge for transfer, as in RT-2 [65], Octo [34], and Pi0.5 [11]. Furthermore, 3D-aware representation learning improves spatial consistency for manipulation [15, 27, 28, 54, 58]. De-

spite these advances, prevailing methods typically assume identical fixed camera configurations during both training and inference, which limits their ability to infer under new camera viewpoints. Our work targets these gaps by pretraining view robust visual representations and enabling flexible view deployment within a unified multitask learner.

2.3. Multi view generation

Recent advances in Multi View Diffusion Model have significantly extended generative modeling from object level synthesis to complex scene level understanding. Early approaches such as Free3D [60] and SV3D [49] explored generating geometrically consistent novel views from a single image without relying on explicit 3D representations. Subsequent works including MV-Adapter [20] and 3D-Adapter [5] introduced geometry aware conditioning mechanisms, enhancing cross view consistency and improving multi view alignment. Building upon these, models like CAT3D [14], and Seva [62] further scaled MVDMs to holistic scene generation, achieving improved spatial coherence and diversity. More recent efforts such as NVComposer [26] and Bolt3D [47] leverage 3D priors to enable rapid, high fidelity multi view synthesis at the scene level. In this work, we are the first to adapt multi-view diffusion models for pretraining view robust representations in

robotic tasks, enabling flexible view deployment and benefit manipulation performance.

3. Method

Overview. In this section, we present our two stage framework for learning 3D consistent visuomotor in details. As illustrated in Fig. 2, DiffuView first pretrains a multi-view diffusion model to infer geometric correspondences across different camera views, enabling the encoder to capture cross view aligned representations. In the second stage, we utilize the pretrained perceptual foundation to extract features, which are aggregated and decoded by a diffusion-based policy network to produce continuous robot actions. Together, these stages allow the robot to learn spatially grounded, viewpoint robust policies for manipulation.

3.1. Multi-View Diffusion Pretraining

Geometric Information Encoding. To enable our model to learn cross view geometric consistency, we use the explicit warp features to control the target view generation. Given a source input $O_i = \{I_i, D_i\}$ captured at pose P_i with intrinsic K_i , we first compute 3D point coordinates. These points are then reprojected into the target camera frame to obtain the warped RGB and depth maps, denoted as $\tilde{I}_{i \rightarrow j}$ and $\tilde{D}_{i \rightarrow j}$:

$$(\tilde{I}_{i \rightarrow j}, \tilde{D}_{i \rightarrow j}) = \text{Warp}(I_i, D_i, P_i, P_j, K_i, K_j). \quad (4)$$

For RGB-D cameras, we use the measured depth D_i directly, while for RGB-only setups, we employ metric depth estimation networks to predict $\tilde{D}_i$, which serves as a substitute for geometric conditioning. Following [4], we further employ dense Plücker ray embeddings to inject pixel-wise camera geometry into the generative process. For each camera i with intrinsic matrix K_i and pose $P_i = [\mathbf{R}_i \mid \mathbf{T}_i] \in \mathbb{R}^{3 \times 4}$, every pixel $u(x, y)$ defines a 3D viewing ray direction and origin as $\mathbf{d}_{i,xy} = \mathbf{R}_i^T K_i^{-1} u$, $\mathbf{o}_j = \mathbf{T}_j$. Each ray is represented in Plücker coordinates:

$$\mathbf{r}_{i,xy} = \langle \mathbf{d}_{i,xy}, \mathbf{m}_{i,xy} \rangle, \quad \mathbf{m}_{i,xy} = \mathbf{o}_i \times \mathbf{d}_{i,xy}, \quad (5)$$

where $\mathbf{d}_{i,xy} \in \mathbb{R}^3$ is the unit direction vector and $\mathbf{m}_{i,xy} \in \mathbb{R}^3$ is its corresponding moment vector. The resulting dense Plücker embedding for camera i is thus

$$\mathbf{e}_{i,xy} = [\mathbf{d}_{i,xy}; \mathbf{m}_{i,xy}] \in \mathbb{R}^6, \quad (6)$$

forming a dense tensor $\mathbf{E}_i \in \mathbb{R}^{6 \times H \times W}$ that encodes per-pixel geometric camera context.

3D Aware Multi View Diffusion Model. The VAE encodes input images into latent tensors of size $\frac{H}{8} \times \frac{W}{8}$. To ensure geometric alignment, we extract corresponding embeddings from the RGB, depth and camera embeddings us-

ing a lightweight convolutional encoder. Then these embeddings are added to the vae latents:

$$\mathbf{z}_{\text{source}} = \mathcal{E}(I_i) + \text{CNN}(I_i, D_i, \mathbf{E}_i), \quad (7)$$

$$\mathbf{z}_{\text{target}} = \mathcal{E}(I_j) + \text{CNN}(\tilde{I}_{i \rightarrow j}, \tilde{D}_{i \rightarrow j}, \mathbf{E}_j), \quad (8)$$

where $\mathcal{E}$ is the VAE encoder. Gaussian noises are added to the target VAE latents $\mathcal{E}(I_j)$ during training, and the model can generate target views via source view control while inference. The Multi View Diffusion Model applies 2D spatial attention within each view to capture intra view spatial dependencies, and employs 3D full attention across all views to reason about cross view geometric relationships. Our model is trained with $N + M = 8$, where N, M mean source and target view numbers respectively. During training, we do not use a fixed number of inputs. Instead, the number of source views N is stochastically varied within $1 \sim 3$. In addition, our model deliberately excludes temporal positional encoding, this design choice prevents the model from memorizing absolute view positions. The benefit is that the model is not bound by the fixed length of the training data, allowing it to naturally extrapolate to an arbitrary number of N and M during inference.

3.2. Policy Learning

Visual Feature Extractor. After pretraining, we repurpose the diffusion UNet as a 3D aware vision encoder capable of extracting spatially consistent representations and performing implicit viewpoint transformation. Thanks to our flexible view inference design, the pretrained model can serve two complementary roles during downstream learning: (i) as a feature extractor for action policy training, and (ii) as a viewpoint-adaptive encoder that enables visuomotor policies to reason from new viewpoints. Similar to the VPP [19], we employ the pretrained diffusion model primarily as a vision encoder rather than a denoiser, by performing only a single diffusion forward pass during inference. Specifically, we extract multi scale features from the UNet upsampling layers, which preserve fine-grained geometric and semantic information. These features are then processed by a FiLM-conditioned Q-Former to obtain compact, task relevant visual embeddings. The FiLM conditioning is guided by the end-of-text (EoT) token from a CLIP text encoder [39], ensuring that the extracted features are modulated by the semantic intent of the language instruction. This design provides a unified interface between pretrained visual representations and downstream action learning, enabling robust visuomotor control.

Diffusion Policy Network. After the FiLM conditioned Q-Former aggregates the visual features into a compact observation embedding $\mathbf{z}_{\text{obs}}$, a diffusion policy is employed as the action head to generate the robot action $\mathbf{a}_0$ conditioned on $\mathbf{z}_{\text{obs}}$ and the language instruction l_{emb} . Following the

standard diffusion process, the forward noising procedure progressively adds Gaussian noise to the clean action $\mathbf{a}_0$:

$$\mathbf{a}^{(t)} = \sqrt{\bar{\alpha}_t} \mathbf{a}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad t \in \{1, \dots, T\}, \quad (9)$$

where $\bar{\alpha}_t = \prod_{k=1}^t (1 - \beta_k)$ denotes the cumulative noise schedule, and $\boldsymbol{\varepsilon}$ is sampled white noise. At each timestep t , the policy network ε_ψ learns to predict the noise component based on the noisy action $\mathbf{a}^{(t)}$, the timestep t , the observation embedding $\mathbf{z}_{\text{obs}}$, and the language token $\mathbf{l}$:

$$\mathcal{L}_{\text{policy}} = \mathbb{E}_{(\mathbf{a}_0, \mathbf{z}_{\text{obs}}, \mathbf{l}), t, \boldsymbol{\varepsilon}} \left[\left\| \boldsymbol{\varepsilon} - \varepsilon_\psi(\mathbf{a}^{(t)}, t, \mathbf{z}_{\text{obs}}, l_{\text{emb}}) \right\|^2 \right]. \quad (10)$$

Following [30], we integrate the observation embedding $\mathbf{z}_{\text{obs}}$ and the language embedding l_{emb} into each diffusion transformer block using cross-attention layers, allowing the policy to attend to both spatial visual context and semantic task instructions. We also replace the standard Layer-Norm with RMSNorm for more stable training. In addition, we introduce an action causal self-attention mechanism to model temporal dependencies among consecutive action tokens. Unlike vanilla self-attention, causal masking enforces an autoregressive constraint such that each action token can only attend to its preceding tokens. This design enables the policy to maintain temporal consistency in sequential action generation, preventing information leakage from future steps and improving smoothness and physical feasibility in multi-step manipulation trajectories. To further enhance efficiency, we adopt a noise timestep conditioned Mixture-of-Experts (MoE) strategy [42]. At each timestep t , the router dynamically activates Top 2 experts within each MoE module on the current noise-level token $\eta(\sigma_t)$, allowing the network to accelerate denoising. During inference, the model iteratively denoises a Gaussian sample to reconstruct the clean action sequence $\mathbf{a}_0$.

4. Experiments

4.1. Pretraining Setups

To enable our pretraining model to generalize effectively to the visual and geometric characteristics of robotic manipulation scenes, we construct a pretraining dataset composed of diverse multi view observations tailored to manipulation tasks. Specifically, we select a subset about 100 tasks of multi view sequences from the RH20T dataset [12], a large scale real world robotic dataset. Each episode contains proprioceptive data collected from multiple calibrated camera viewpoints using various robot arms and grippers. Since the original RH20T depth maps are often noisy and unreliable, we employ MapAnything [23] to estimate metric depth maps. Furthermore, the camera configurations in RH20T are relatively fixed, so we additionally generate diverse robot centric RGB-D observations in simulation environments. Specifically, we render the data using both the

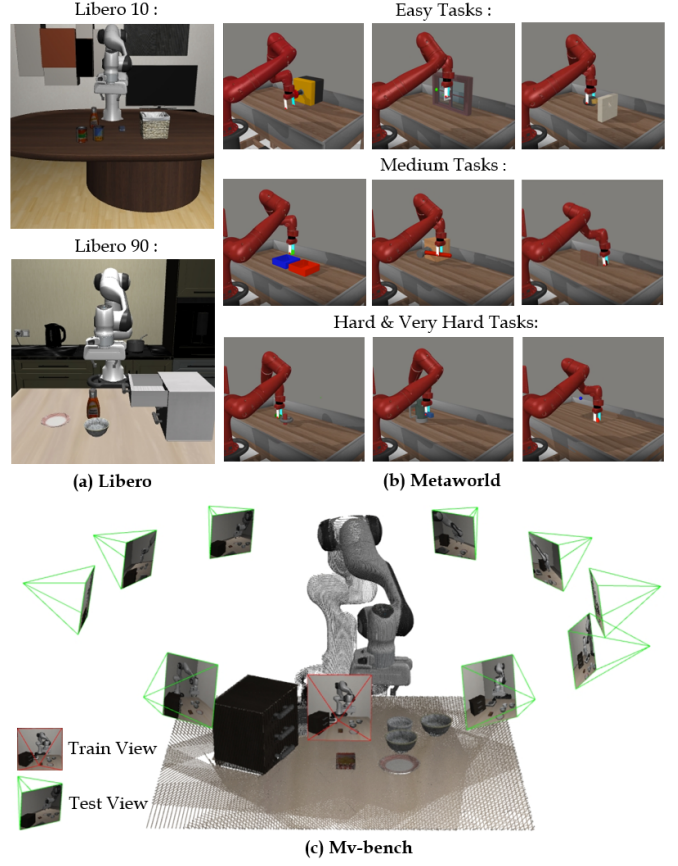


Figure 3. Simulation benchmarks. We conducted our experiments on two established benchmarks, LIBERO and MetaWorld, as shown in (a) and (b), respectively. These benchmarks collectively encompass a diverse array of manipulation tasks with varying levels of difficulty and complexity. Furthermore, we also conducted experiments on our self-formulated benchmark, Mv-Bench, which is specifically designed to evaluate the view adaptive capability of our method, as illustrated in (c).

RoboSuite and CoppeliaSim simulators, where over 5,000 camera viewpoints are randomly sampled around robotic manipulation scenes to capture diverse spatial perspectives. Our fine-tuning dataset encompasses over 200 robotic manipulation tasks with rich and varied object interactions. All RGB-D images are resized and standardized to a resolution of 512×512 during training. We fine-tune our multi view diffusion model on these robot-centric datasets using 8 NVIDIA A100 GPUs, using about 2 days. This design improves the capability of the pretrained model for downstream visuomotor learning and provides strong generalization between viewpoints and task types.

4.2. Simulation Experiments

Benchmarks. We evaluate our proposed method on two widely adopted robotic manipulation benchmarks: Libero [29] and MetaWorld [56]. The Libero benchmark

Table 1. Quantitative results on Libero Benchmark.

Benchmark	DP [6]	MT-ACT [1]	MDT [44]	OpenVLA [25]	Dita [18]	VQVLA [50]	$\pi_{0.5}$ -ki [11]	DiffuView (Ours)
Libero 10	51.0	68.0	65.0	53.7	63.8	-	84.8	89.2
Libero 90	69.0	54.0	66.9	74.8	-	81.0	92.7	92.5
Average (100 tasks)	67.2	55.4	66.7	72.7	-	-	91.9	92.2

Table 2. Quantitative results on MetaWorld Benchmark.

Tasks Level (Numbers)	RT-1 [3]	DP [6]	SuSIE [2]	GR-1 [51]	Baku [16]	VPP [19]	DiffuView (Ours)
Easy (28 tasks)	0.605	0.442	0.560	0.725	0.831	0.818	0.836
Medium (11 tasks)	0.042	0.062	0.196	0.327	0.527	0.493	0.542
Hard&Very Hard (11 tasks)	0.015	0.095	0.255	0.451	0.346	0.526	0.537
Average (50 tasks)	0.351	0.282	0.413	0.577	0.530	0.682	0.706

provides a comprehensive suite of simulated manipulation tasks designed to assess spatial reasoning and generalization capabilities in visuomotor learning. In particular, we focus on the Libero-90, Libero-10 and Libero-Spatial subsets, which include a diverse set of manipulation scenarios. Each task contains 50 demonstrations using a Franka Panda robot, and we follow the standard evaluation protocol established in prior work [42, 43]. In addition, we evaluate DiffuView on the MetaWorld benchmark, which is based on the MuJoCo Simulator. Following the difficulty categorization defined in [45], we consider all 50 tasks grouped into three levels: easy, medium, and hard & very hard. We collect 50 trajectories for each task in the Meta-World benchmark as our training dataset. The observation inputs are third person RGB-D images, while the action space includes the end-effector position and gripper state.

Comparison Methods. To rigorously evaluate the effectiveness of our proposed method, we conduct quantitative comparisons against several state-of-the-art methods featuring distinct model structures. Specifically RT-1 [3], MT-ACT [1] and Baku [16] are transformer based multi task imitation learners. DP [6], Dita [18], and MDT [43] are diffusion based multi task action generators. Notably, both Dita and MDT employ causal self-attention within the network architecture, a design choice shared by our method. Additionally, we compare against OpenVLA [25], VQVLA [50], and $\pi_{0.5}$ -ki [11], which constitute representative VLA models. Finally, SuSIE [2], GR-1 [51] and VPP [19] leverage visual generation tasks to enhance the effectiveness of action learning. SuSIE conditions policies on future goal images synthesized via image editing diffusion models, GR-1 autoregressively generates both future frames and actions in one framework, and VPP introduces predictive visual representations derived from video diffusion models to guide policy learning with explicit future dynamics.

Implementation Details. For the noise-level conditioning module, we employ a MoE design within each transformer

Table 3. Mv-bench results.

Angles	-30°	-15°	0°	15°	30°	45°	60°	Average
OpenVLA [25]	25.7	62.4	84.7	54.8	26.4	12.6	8.2	39.3
DiffuView (Ours)	52.4	67.9	86.2	72.9	55.3	44.9	34.6	59.2

block, consisting of 4 experts, among which Top-2 experts are activated during inference. To perform the denoising process, we utilize the DDIM-solver, a numerical ODE-based sampler designed for diffusion models, and apply 10 denoising steps during inference. In our multi task action learning framework, we use 8 layers transformer blocks with a latent dimension of 768. We use 2 observation frames at one viewpoint and language embeddings from CLIP as inputs then output an action chunk size of 10. All experiments are conducted on 8 NVIDIA A100 GPUs. The results are shown in Tab. 1 and Tab. 2.

4.3. View Generalization Experiments.

Following [59], we also render multi-view data in the LIBERO Spatial environment to evaluate the robustness of our method under viewpoint variations. Our policy model only takes one view’s feature as input during both training and inference. However, our pretrained model serves dual roles — acting as a feature extractor during the training phase, and as a view-adaptive module during inference. To assess the view adaptive capability of our pretrained module, we train the policy using data from one fixed view (agent view in Libero) and perform inference across a large set of unseen viewpoints in the same environment. These viewpoints are generated by rotating the agent view camera around the robot base coordinate system along the z-axis by fixed angular intervals, forming a new benchmark, which we denote as Mv-Bench for evaluating cross view generalization. As summarized in Tab. 3, the pretrained module enables the policy to maintain stable manipulation performance under large viewpoint shifts, whereas the baseline

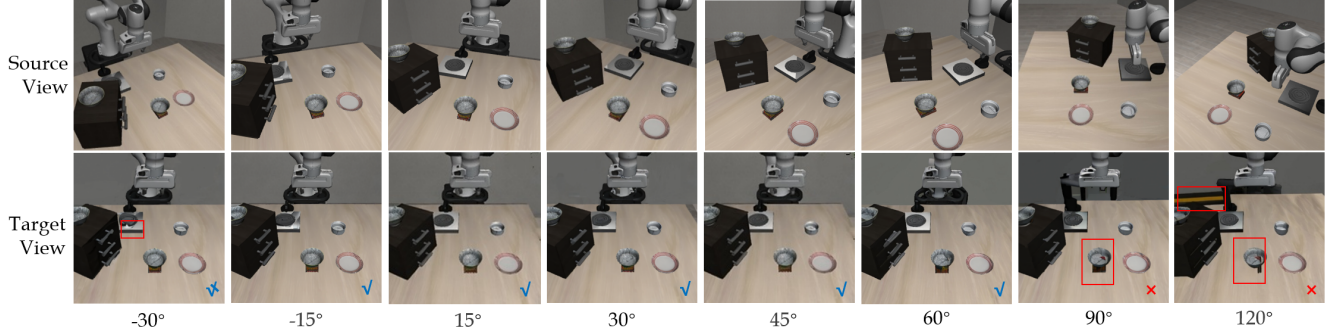


Figure 4. This figure illustrates the effect of our pretrained model serving as a view-adaptive module. The results herein demonstrate that our approach competently handles view transformation within $-30^\circ \sim 60^\circ$.

Table 4. Real world experiment results on success rate.

Methods	Wipe table	Return bowl	Open drawer	Pull out issue	Average
DP [6]	0.28	0.49	0.75	0.51	0.51
OpenVLA [25]	0.49	0.62	0.74	0.68	0.63
DiffuView (Ours)	0.68	0.60	0.77	0.54	0.65

method struggles when the camera deviates slightly from the training angle. While Tab. 3 provides quantitative evidence of our model’s robustness, Fig. 4 offers a qualitative comparison of our pretrained model’s capability of view transformation. However, when the viewpoint shift becomes excessively large, spatial geometric occlusions occur, leading to a noticeable degradation in the performance of the pretrained model. In this setting, specifically where the number of source views is set to $N = 1$ and target views to $M = 1$, the results further validate the highly flexible view deployment capability of our pretrained model.

4.4. Real World Experiments

Experimental setup. In our real-world experiments, we conducted robotic manipulation tasks using the Franka Research 3 robotic arm, as illustrated in Fig. 5 (right). The setup included multiple cameras, including RealSense D435 and D455, to provide various viewpoints and sensor configurations for the robotic system to process.

Real world tasks. We demonstrate the effectiveness of our proposed method in 8 typical manipulation scenarios. These tasks, as illustrated in Fig. 5 (left), involve a range of activities that showcase the robot’s ability to handle diverse objects and interactions. The tasks include collect bowls placed randomly on the table and arrange them in a stacked position; pick up a sponge and clean the table; pick up a kettle and water plants; open a drawer; pick up small fruits, such as lychees, and place them into a designated bowl; extract a single tissue from the box; pick up a soft toy, like a doll, and place it on the table. These tasks were designed to test the robustness of the learned policy under varying visual conditions and object interactions. Each task was collected

Table 5. Ablation Results.

Ablation Types	Success Rate
DiffuView	89.2
DiffuView w/o Robotics Data Pretraining	63.3
DiffuView w/o Plücker Embedding	76.2
DiffuView w/o FiLM Conditioning in Q-Former	73.3
DiffuView Noise Conditioned Activated Experts Top $K = 1$	87.7

with 50 trajectories with randomized object positions and distractor objects to ensure robustness.

Quantitative results. As shown in Tab. 4, we select four particularly challenging real-world manipulation tasks to demonstrate the effectiveness of our approach. The results indicate that our method significantly improves generalization, with the DiffuView framework achieving superior performance compared to prior models. Our approach surpasses traditional 2D-based methods, showcasing its robustness. The real-world results validate that incorporating 3D-consistent visual representations provides substantial benefits for manipulation, making the proposed method highly effective across diverse environments and task settings.

4.5. Ablation Studies

Based on the above experiments, we raise the following questions: **(I)** Is it effective to pretrain the multi-view diffusion model in a robot-centric setting? **(II)** Is the injection of Plücker embedding information beneficial for improving spatial representation? **(III)** When aligning the pretrained visual features to the policy network through the Q-Former, is the task-relevant FiLM conditioning effective? **(IV)** How does the number of activated experts in the noise-conditioned MoE affect the policy performance? **(V)** How well does the multi view diffusion model perform in real-world robotic environments? **(VI)** Can the pretrained multi view diffusion model also generalize to wrist view observations, beyond the canonical camera viewpoints?

To better understand the contribution of each component in our framework, we conduct ablation studies on

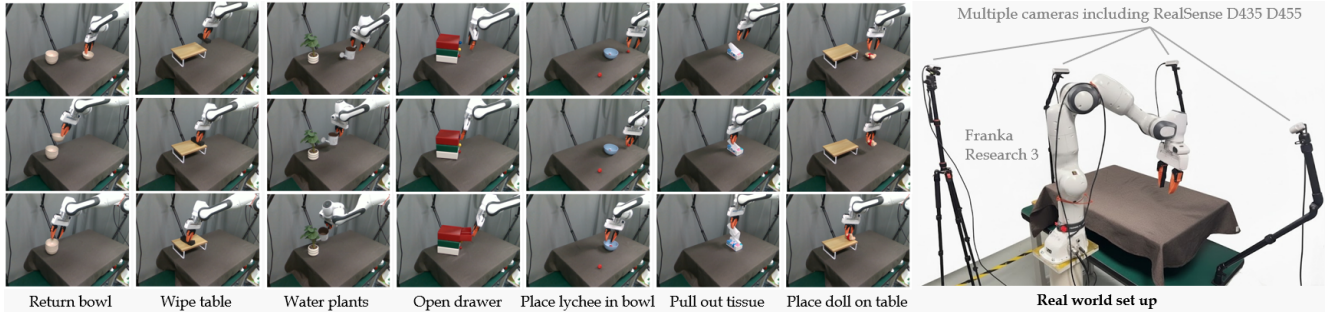


Figure 5. Successful rollouts of DiffuView (left) and real world set-up (right). The image illustrates the robotic arm performing various manipulation tasks, showcasing the real-time execution of the DiffuView model.

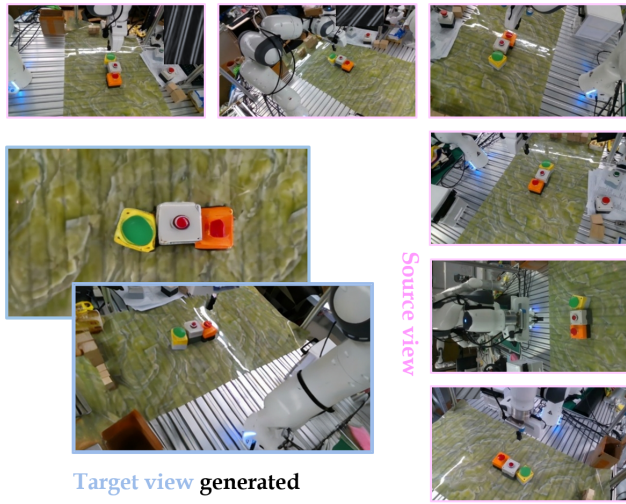


Figure 6. Wrist view and third view generation in real-world dataset. The pink bounding boxes delineate the source images captured from various viewpoints, while the blue bounding box indicates the target image requiring view restoration.

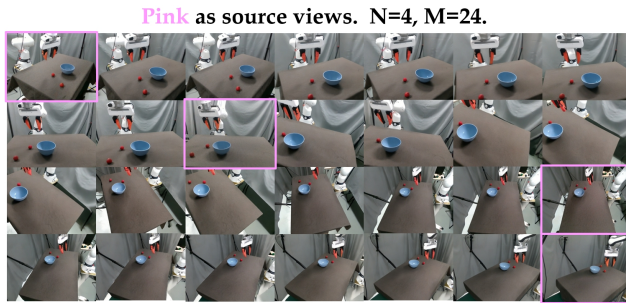


Figure 7. Qualitative Results of Real-World Multi-View Diffusion.

the Libero 10 benchmark. As shown in Tab. 5, when the model is trained without robot centric data, the performance drops sharply, highlighting the importance of pretraining on robot centric datasets. Removing the Plücker embedding also leads to a noticeable decline, confirming that in-

jecting explicit geometric camera ray information enhance our method. Furthermore, excluding the FiLM-based language conditioning in the Q-Former reduces the success rate to 73.3%, indicating that task relevant modulation is crucial for aligning pretrained visual representations with the downstream policy. Lastly, when we only activate only 1 expert of the noise conditioned MoE, the performance decreases to 87.7%. In addition, as shown in Fig. 6, although our model is not trained on wrist view images, it can still successfully generate results. The Fig. 7 presents the model’s generation results in real-world settings, while also showcasing the adaptive scheme for varying numbers of source and target views during inference. Since features extracted from different camera viewpoints follow different distributions, this setting essentially corresponds to an out-of-distribution generalization problem. Nevertheless, the strong mapping and generative capability of the diffusion model alleviates this issue to a large extent, providing an effective view adaptive module that generalizes well to unseen viewpoints such as the wrist view.

5. Conclusion and Limitation

In this work, we proposed DiffuView, a two stage framework that learns 3D consistent visuomotor representations through multi-view diffusion pretraining and multi-task diffusion based policy learning. Our approach achieves state-of-the-art performance across multiple simulated manipulation benchmark. Furthermore, we evaluated the viewpoint generalization metrics on proposed MV-bench, confirming that our work can robustly handle large viewpoint shifts. These results highlight the effectiveness of our pretraining in providing a new paradigm for representation learning in robotic manipulation. Although our pretraining framework supports flexible view inference and enables viewpoint robust policy learning, it currently lacks an explicit understanding of dynamic temporal information. This limits the model’s ability to reason about motion continuity dependencies. In future work, we plan to extend DiffuView toward a joint flexible view and time pretraining framework, enabling unified spatial temporal representation learning.

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